Prototype a Grounded Assistant

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected grounded-prototype record.

Explain stages, trace mismatched evidence, audit a happy-path claim and design a revealing test.

Keep one habit: Keep inputs, eligible sources, draft, validation, final status and remaining gaps together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Block to Python Bridge. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Tools and APIs. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Retrieval Augmented Generation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Block to Python Bridge

Tools and APIs

Retrieval Augmented Generation

W1 · Explain the prototype stages

> Give the purpose and evidence for each stage.

Your evidence record

- Room W2: request B, approval B, register C capacity10; same-room support fails, needs_review/null claims. Room contract requires current same-room approval+register, nonnegative integer capacity and matched sources; read-only. Separate plant prototype: PL2-v1 seedlings need water; PL3-v1 this activity checks soil daily. Daily-check answer supported only by PL3.

  Missing PL3 or an unsupported watering schedule must not become an accepted answer.

- Reference for this exercise: open this week's student lesson, "Prototype a Grounded Assistant", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Trace the room mismatch

> Approval for B and capacity for C cannot establish B capacity.

Your evidence record

- Room W2: request B, approval B, register C capacity10; same-room support fails, needs_review/null claims. Room contract requires current same-room approval+register, nonnegative integer capacity and matched sources; read-only. Separate plant prototype: PL2-v1 seedlings need water; PL3-v1 this activity checks soil daily. Daily-check answer supported only by PL3.

  Missing PL3 or an unsupported watering schedule must not become an accepted answer.

- Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Audit a happy-path guarantee

> One correct path does not demonstrate fallback and access behavior.

Your evidence record

- Room W2: request B, approval B, register C capacity10; same-room support fails, needs_review/null claims. Room contract requires current same-room approval+register, nonnegative integer capacity and matched sources; read-only. Separate plant prototype: PL2-v1 seedlings need water; PL3-v1 this activity checks soil daily. Daily-check answer supported only by PL3.

  Missing PL3 or an unsupported watering schedule must not become an accepted answer.

- A report omits this qualification: A correct happy path does not establish **fallback**, access control or side-effect safety. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Design a revealing stage failure

> Supply complete inputs and a predicted first failure before trying the prototype.

Your evidence record

- Room W2: request B, approval B, register C capacity10; same-room support fails, needs_review/null claims. Room contract requires current same-room approval+register, nonnegative integer capacity and matched sources; read-only. Separate plant prototype: PL2-v1 seedlings need water; PL3-v1 this activity checks soil daily. Daily-check answer supported only by PL3.

  Missing PL3 or an unsupported watering schedule must not become an accepted answer.

- Reference for this exercise: open this week’s student lesson, “Prototype a Grounded Assistant”, at “See how it works” and “Follow a worked example”. Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Trace the new zero-capacity room response independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Apply the separate room contract: supported requires current matching approval and register records for the requested room, a valid nonnegative integer capacity and correctly referenced sources. Otherwise needs_review with null claims. Read-only: no booking. Use only the given independent records and draft; no new room measurement is provided.

Request room J; approval and register both J capacity 0. Draft is well-formed with J/0 and correct sources. Trace stages and distinguish zero from missing.

My independent explanation, trace and limits

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Apply a new case independently

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Connect user interface, source store, retrieval, prompt, schema validation, and **evidence** display.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Capstone grounded assistant. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Capstone grounded assistant. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Build a paper assistant from notes A='Roots take up water' and B='Leaves use light'. Answer 'Which part takes up water?' and 'How tall is this plant?' with references or **fallback**.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Approved notes say plants need light and roots take up water. Ask how roots help.

Specify the minimum grounded path.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A study assistant has a supplied passage about roots. A user asks a question about planets. What should the **prototype** do within a passage-only boundary?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 24 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Build a paper assistant from notes A='Roots take up water' and B='Leaves use light'. Answer 'Which part takes up water?' and 'How tall is this plant?' with references or **fallback**.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

Request room J; approval and register both J capacity 0. Draft is well-formed with J/0 and correct sources. Trace stages and distinguish zero from missing.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

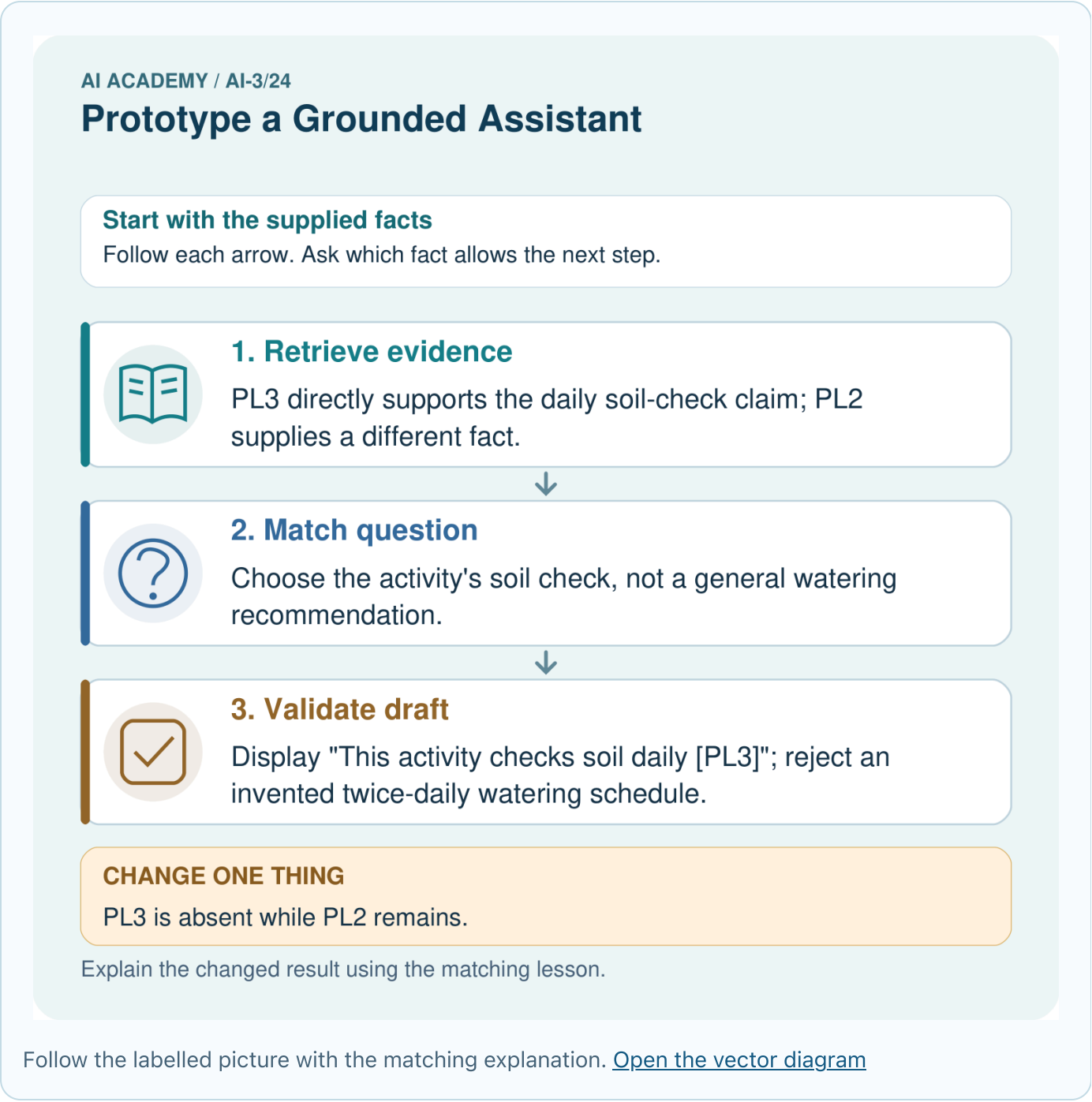
> Keep the explanation and the support you used.

Use this when helpful

Build a minimal grounded study answer with checks between retrieval and display.

Source PL2: seedlings need water. Source PL3: this activity checks soil daily. Question: what does this activity check daily?

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Build a minimal grounded study answer with checks between retrieval and display.
- Which labelled step matches this detail? Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule.
- 1. Retrieve **evidence**

- 2. Match question
- 3. Validate draft
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Validate draft Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. Read the picture in order.

  Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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> | |
> |---|
> | |
> | |
> | |

▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: PL3 is absent while PL2 remains. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. Explain that the daily check is not specified in the available source.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Explain that the daily check is not specified in the available source. The remaining general water fact does not establish this activity's schedule.

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review Block to Python Bridge (ai-3/23). Goal: Translate a short paper calculation into a Python script. Short practice: Trace the changed list and identify where the result differs.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Tools and APIs (ai-3/21). Goal: Explain the **mechanism** and terminology of tools and api contracts. Short practice: Does a valid C7 ID permit the extra lending action?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Retrieval Augmented Generation (ai-3/18). Goal: Explain the **mechanism** and terminology of building a grounded answer. Short practice: Which part of the source supports a two-minute melting claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
>
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> |---|---|
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> | | |
> | | |

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Prototype a Grounded Assistant: the case

AI ACADEMY / AI-3/24 · 01 · LOOK AT THE FACTS

Prototype a Grounded Assistant: the case

The situation

Build a minimal grounded study answer with checks between retrieval and display.

Supplied evidence

Source PL2: seedlings need water. Source PL3: this activity checks soil daily. Question: what does this activity check daily?
All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

By the end, I can explain prototype a grounded assistant using a traced example and a stated limitation.

Follow the labelled picture with the matching explanation. Open the vector diagram

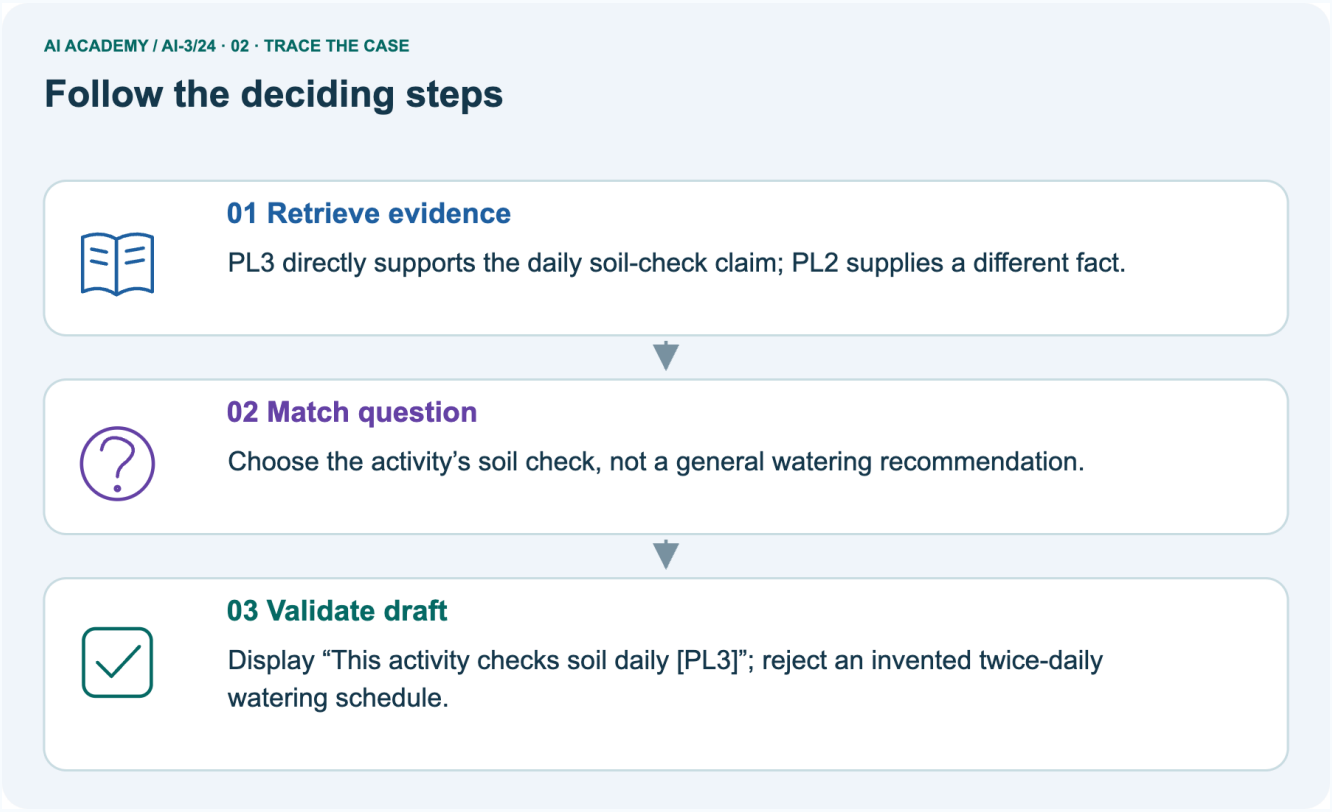
Build a minimal grounded study answer with checks between retrieval and display. Supplied facts: Source PL2: seedlings need water. Source PL3: this activity checks soil daily. Question: what does this activity check daily?

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



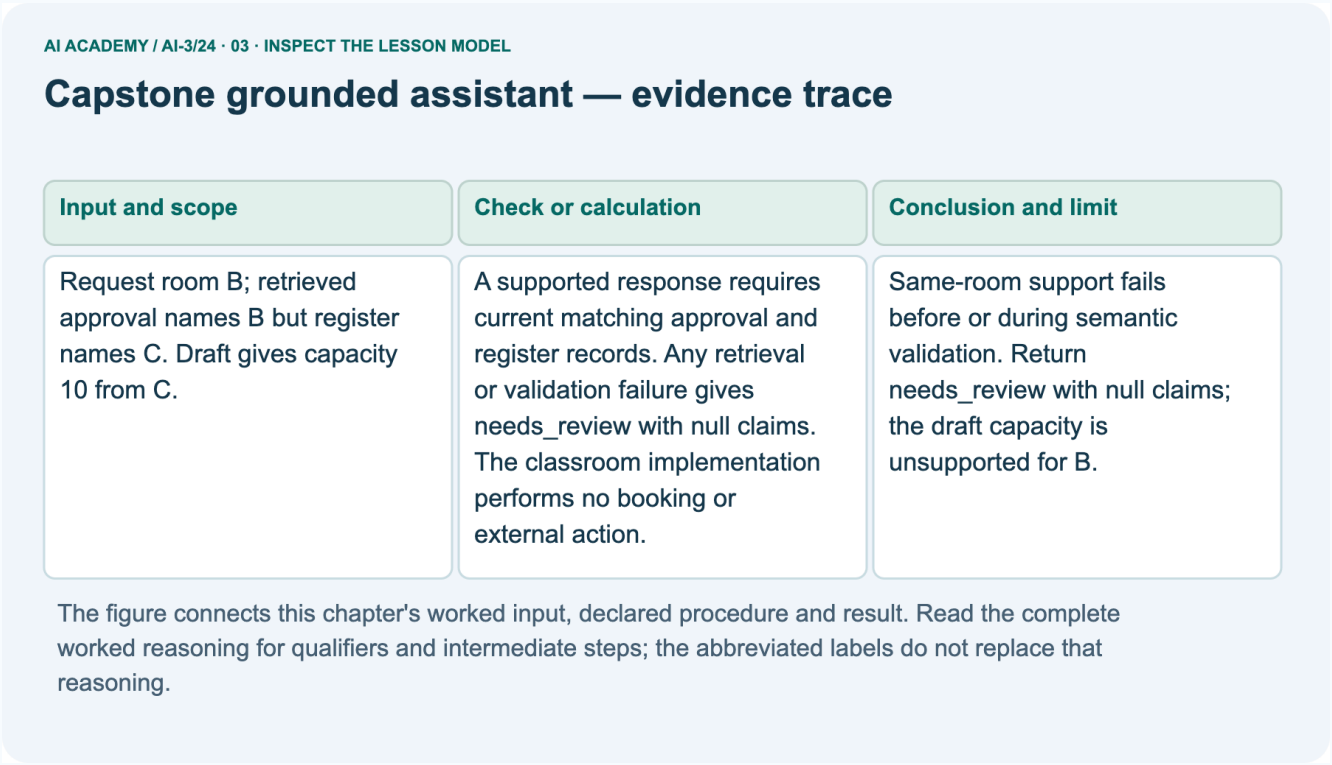
Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve **evidence**: PL3 directly supports the daily soil-check claim; PL2 supplies a different fact. → Match question: Choose the activity's soil check, not a general watering recommendation. → Validate draft: Display "This activity checks soil daily [PL3]"; reject an invented twice-daily watering schedule.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone grounded assistant — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone grounded assistant — **evidence** trace. Request room B; retrieved approval names B but register names C. Draft gives capacity 10 from C. | A supported response requires current matching approval and register records.

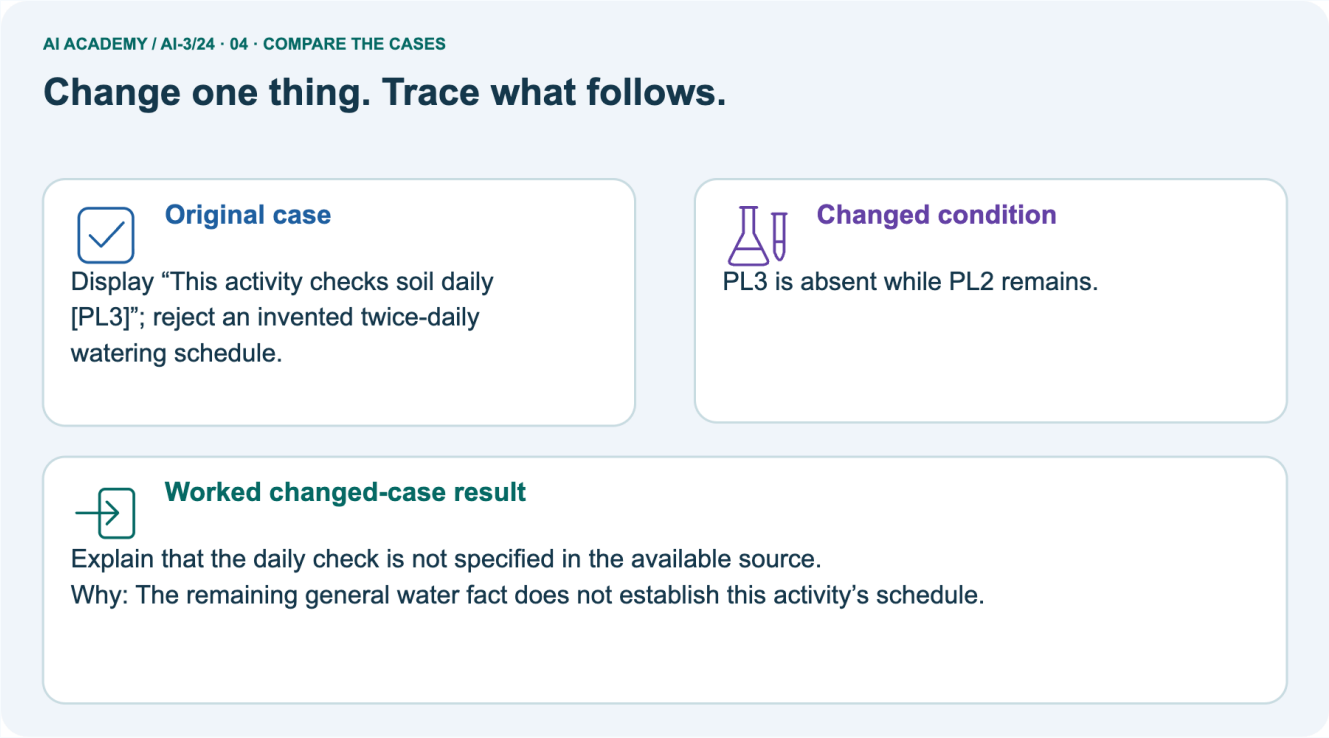
Any retrieval or validation failure gives needs_review with null claims. The classroom implementation performs no booking or external action. | Same-room support fails before or during semantic validation.

Return needs_review with null claims; the draft capacity is unsupported for B.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

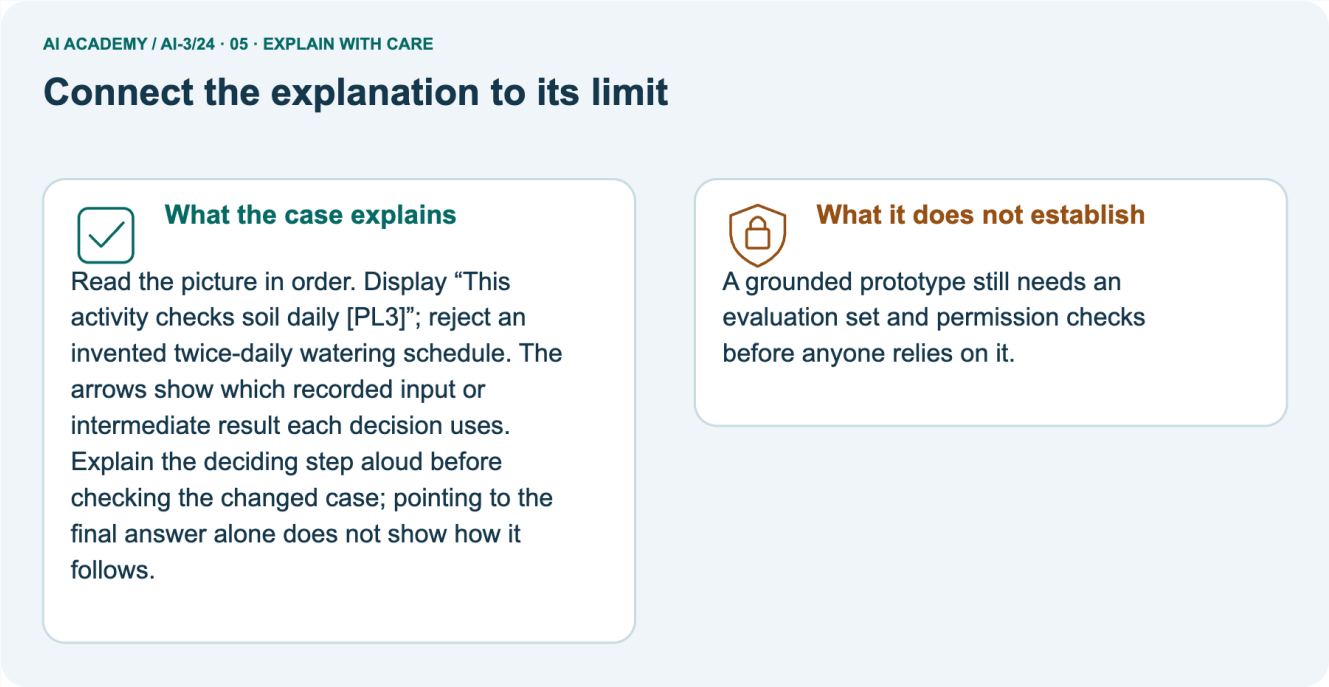
Original case: Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. Changed condition: PL3 is absent while PL2 remains.

Expected result: Explain that the daily check is not specified in the available source. Reason: The remaining general water fact does not establish this activity’s schedule.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Display “This activity checks soil daily [PL3]”; reject an invented twice-daily watering schedule. The arrows show which recorded **input** or intermediate result each decision uses.

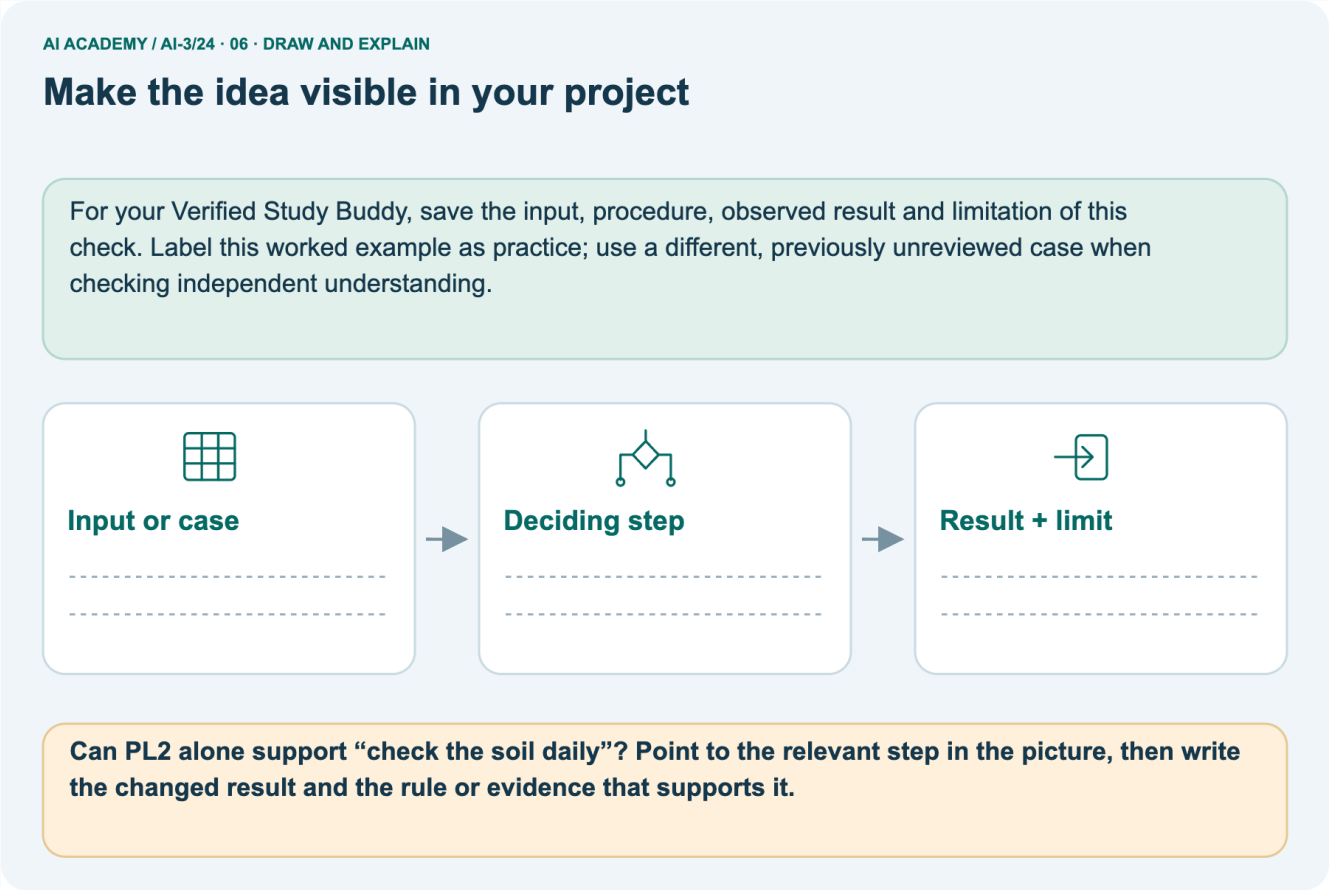
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A grounded **prototype** still needs an evaluation set and permission checks before anyone relies on it.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can PL2 alone support “check the soil daily”? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

01. Prototype a Grounded Assistant: the case

02. Follow the deciding steps

03. Capstone grounded assistant — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project
</details>

<details>
<summary>Optional interactive practice and completed work</summary>

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.
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